

- A **pie chart** allows us to compare the popularity of different categories by comparing the sizes of slices of a pie. For example, we can see that blackforest cake made up one quarter of the cake sales, and that lemon cake was the least popular choice.



The **mode** of a set of categorical data is the category which occurs most often.

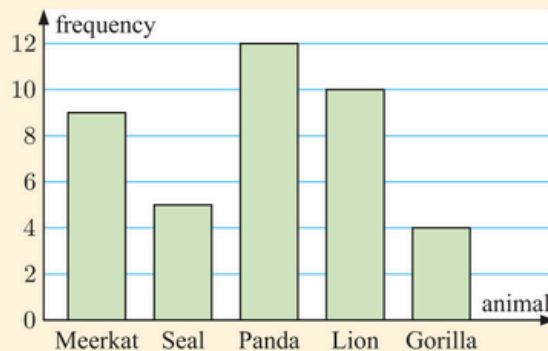
In this case, the mode of the data is *chocolate cake*.

Example 2

Self Tutor

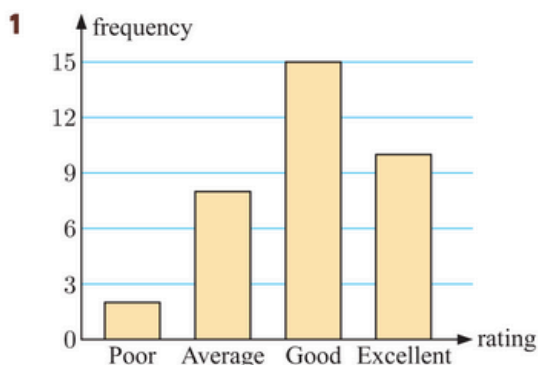
40 zoo visitors were asked to name their favourite animal. Their responses are shown on the column graph.

- How many people liked the lion the most?
- What percentage of people liked the seal the most?
- Find the mode of the data.



- 10 people liked the lion the most.
- $\frac{5}{40} \times 100\% = 12.5\%$ of people liked the seal the most.
- The mode of the data is *panda*.

EXERCISE 14B

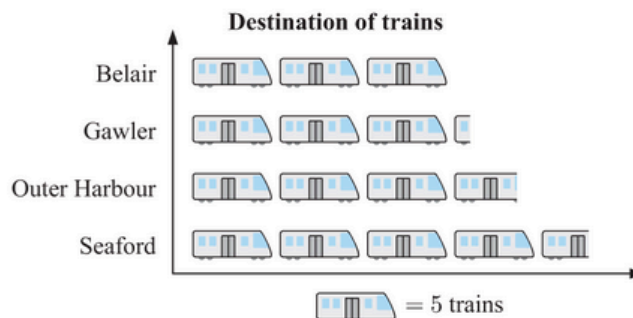


A hotel manager wanted feedback from her guests about the hotel's facilities. She asked the guests to rate the facilities, and received the responses shown on the column graph.

- How many guests were surveyed?
- Classify the data collected.
- How many guests rated the facilities as 'Average'?
- Find the mode of the data.
- What percentage of the guests rated the facilities as 'Good' or better?

- 2** This picture graph shows the destinations of trains leaving the Adelaide train station one morning.

- a** How many trains were travelling to:
- Belair
 - Outer Harbour?
- b** Find the mode of the data.



3



This pie chart shows the items borrowed from a library in one hour.

- a** Find the mode of the data.
- b** Is it possible to determine the number of DVDs borrowed? Explain your answer.
- c** Are the following statements true or false?
- The majority of the items borrowed were fiction books.
 - Non-fiction books made up approximately one quarter of the items borrowed.

- 4** The data below shows the favourite positions of 50 netball players.

GS $\equiv$ Goal Shooter, GA $\equiv$ Goal Attack, WA $\equiv$ Wing Attack, C $\equiv$ Centre, WD $\equiv$ Wing Defence, GD $\equiv$ Goal Defence, and GK $\equiv$ Goal Keeper.

GS	GA	C	WD	WA	GS	C	GK	GA	WA
C	GD	WA	C	GS	GA	C	GD	WA	GD
GS	GK	GS	GD	GA	GK	WD	GS	GK	C
C	GA	C	WD	C	GS	GA	C	GS	GA
WA	GA	GD	C	GS	WA	C	GD	GK	WA

- a** Use a tally and frequency table to organise the data.
- b** Use a column graph to display the data.
- c** Find the mode of the data.
- d** What percentage of players named one of the wing positions as their favourite?
- 5** Courtney filled her drinks machine with bottles of cola, lemonade, water, and orange juice. When she returned a week later to refill the machine, she found that 20 cola, 12 lemonade, 8 water, and 10 orange juice bottles had been sold.

- a** Draw a picture graph to display the number of each type of drink sold for the week.

Use  to represent 5 bottles.

- b** Find the mode of the data.
- c** What percentage of the drinks sold were either water or orange juice?



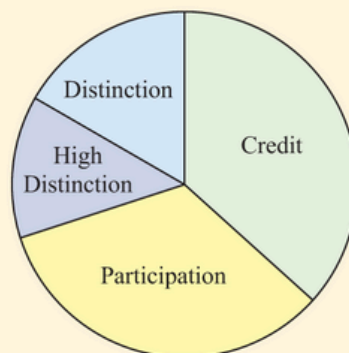
Example 3**Self Tutor**

The results obtained by a class of students in a national Mathematics competition are shown alongside.

- Draw a pie chart to display the data.
- Find the mode of the data.
- What percentage of the students scored a Distinction or better?

Result	Frequency
High Distinction	4
Distinction	5
Credit	11
Participation	10
Total	30

- There are 30 students in total.
Each student represents $\frac{1}{30}$ th of the pie chart, which is $\frac{1}{30} \times 360^\circ = 12^\circ$.
The sector angles for each category are:
High Distinction: $4 \times 12^\circ = 48^\circ$
Distinction: $5 \times 12^\circ = 60^\circ$
Credit: $11 \times 12^\circ = 132^\circ$
Participation: $10 \times 12^\circ = 120^\circ$



- The mode of the data is *Credit*.
- $\frac{4+5}{30} \times 100\% = 30\%$ of the students scored a Distinction or better.

- A termite inspector surveyed 40 of his customers to find out how they heard about him. The responses are given alongside.

- Draw a pie chart to display the data.
- Find the mode of the data.

Response	Frequency
Internet	24
Newspaper	11
Radio	5

- Did internet advertising generate more customers than newspaper and radio combined?

- Lewis recorded the colours of the cars which passed by his house. The results are shown below using W $\equiv$ White, B $\equiv$ Black, b $\equiv$ Blue, R $\equiv$ Red, Y $\equiv$ Yellow, and S $\equiv$ Silver.

W B S W S R b W W Y S W B S W
S W S R B W W R B S W Y W W b
b S B W Y S B S W b S W W B R
S W W R b B S W S Y W b W W S

- Use a tally and frequency table to organise the data.
- Draw a pie chart to display the data.
- Are the following statements true or false?
 - Over one third of the cars were white.
 - Over one quarter of the cars were red or blue.

Discussion

When displaying categorical data, in which situations is it most appropriate to use:

- a column graph
- a picture graph
- a pie chart?